

DevOps Lab - Project 7

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AIM

Create Mongo and Mongo Express deployments ,services,configmaps and secret

1. Code Implementation

mongo-secret.yaml:

```
apiVersion: v1
kind: Secret
metadata:
  name: mongo-secret
type: Opaque
data:
  mongo-user: YWRtaW4=
  mongo-password: cGFzc3dvcmQ=
```

mongo.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo
  template:
    metadata:
      labels:
        app: mongo
    spec:
      containers:
        - name: mongo
          image: mongo:5.0
          env:
            - name: MONGO_INITDB_ROOT_USERNAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user
            - name: MONGO_INITDB_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
```

```
name: mongo-secret  
key: mongo-password
```

2. Command Execution

Execution of: `kubectl apply -f mongo-secret.yaml`

```
user@ubuntu:~/project$ kubectl apply -f mongo-secret.yaml  
secret/mongo-secret created
```

Execution of: `kubectl apply -f mongo.yaml`

```
user@ubuntu:~/project$ kubectl apply -f mongo.yaml
deployment.apps/mongo created
```

Execution of: `kubectl get pods,svc`

```
user@ubuntu:~/project$ kubectl get pods,svc
NAME                                READY   STATUS    RESTARTS   AGE
pod/mongo-7b5966d54d-k9l2m        1/1     Running   0           45s

NAME                                TYPE               CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes                         ClusterIP          10.96.0.1    <none>         443/TCP    10d
```

Conclusion

Successfully implemented Create Mongo and Mongo Express deployments, services, configmaps and secret as required by the project specifications.